

Supervised Bayesian Matrix Factorization

Corrected Update Derivations

Revision of the February 12, 2026 Document

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March 2026

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1 Notation and Model Setup

1.1 Index Conventions

Symbol	Dimension	Meaning
n	scalar	number of samples (patients)
p	scalar	number of features (genes / CpG sites)
K	scalar	number of latent factors
i	$1 \dots n$	sample index
j	$1 \dots p$	feature index
k, k'	$1 \dots K$	factor index
Y	$n \times p$	observed genomics data matrix
L	$n \times K$	loading matrix; l_{ik} = loading of sample i on factor k
F	$p \times K$	factor matrix; f_{jk} = weight of feature j on factor k
β	$K \times 1$	survival coefficient vector
τ	$n \times p$	precision matrix (inverse noise variance)
(t_i, δ_i)		survival time and event indicator for sample i

1.2 Posterior Moment Notation

Throughout these derivations, $q(\cdot)$ denotes the variational posterior and $g(\cdot)$ denotes the prior. We distinguish carefully between the *posterior mean* and the *posterior second moment*:

$$\bar{l}_{ik} := \mathbb{E}_q[l_{ik}] \quad (\text{posterior mean}) \quad (1)$$

$$\overline{l_{ik}^2} := \mathbb{E}_q[l_{ik}^2] = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2 \quad (\text{posterior second moment}) \quad (2)$$

and analogously $\bar{f}_{jk}$, $\overline{f_{jk}^2}$, $\bar{\beta}_k$, $\overline{\beta_k^2}$. These two quantities are **not** equal whenever there is posterior uncertainty. This distinction is essential for the precision update (Section 7) and for the “error-in-variables” correction in the β update (Section 6).

1.3 Models

Revision Note

[R1] The 2/12/26 document wrote $Y = L_{n \times j} F'_{n \times k} + E_{k \times j}$, which has inconsistent and impossible dimensions. Corrected below.

Genomics model.

$$Y_{n \times p} = L_{n \times K} F_{K \times p}^\top + E_{n \times p}, \quad E_{ij} \sim \mathcal{N}(0, \tau_{ij}^{-1}) \quad (3)$$

Element-wise: $Y_{ij} = \sum_{k=1}^K l_{ik} f_{jk} + \varepsilon_{ij}$.

Survival model (Cox proportional hazards).

$$h(t_i) = h_0(t_i) \exp(\eta_i), \quad \eta_i = \mathbf{l}_i^\top \beta = \sum_{k=1}^K l_{ik} \beta_k \quad (4)$$

where h_0 is an unspecified baseline hazard.

Hierarchical priors.

$$l_{ik} \stackrel{\text{iid}}{\sim} g_l \in \mathcal{G}_l, \quad f_{jk} \stackrel{\text{iid}}{\sim} g_f \in \mathcal{G}_f, \quad \beta_k \stackrel{\text{iid}}{\sim} g_\beta \in \mathcal{G}_\beta$$

All three priors are learned from data via the Empirical Bayes Normal Means (EBNM) framework wang2022.

2 Objective Function (ELBO)

The Evidence Lower Bound (ELBO) to be *maximized* over posteriors (q_l, q_f, q_β) and priors (g_l, g_f, g_β) and precision τ is

$$\begin{aligned} \mathcal{F} = & \mathbb{E}_{q_l, q_f, q_\beta} [\log P(Y, t, \delta \mid L, F, \beta, \tau)] \\ & + \sum_k \mathbb{E}_{q_{l_k}} \left[\log \frac{g_{l_k}(l_k)}{q_{l_k}(l_k)} \right] + \sum_k \mathbb{E}_{q_{f_k}} \left[\log \frac{g_{f_k}(f_k)}{q_{f_k}(f_k)} \right] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \end{aligned} \quad (5)$$

The joint log-likelihood decomposes as (conditional independence of Y and (t, δ) given L):

$$\log P(Y, t, \delta \mid L, F, \beta, \tau) = \underbrace{\log P(Y \mid L, F, \tau)}_{\text{genomics}} + \underbrace{\log P(t, \delta \mid L, \beta)}_{\text{survival}} \quad (6)$$

3 Taylor Approximation of the Cox Partial Likelihood

Because $\log P(t, \delta \mid L, \beta)$ is non-conjugate in L and β , we use a 2nd-order Taylor expansion to obtain a locally-Gaussian surrogate that can be handled by EBNM.

3.1 The Working Response

Let $\hat{\eta}_i$ be the current expansion point. The 2nd-order Taylor expansion around $\hat{\eta}$ gives

$$\ell(\eta) \approx \ell(\hat{\eta}) + \sum_i (\eta_i - \hat{\eta}_i) u_i + \frac{1}{2} \sum_i (\eta_i - \hat{\eta}_i)^2 (-W_{ii}) \quad (7)$$

where

$$u_i = \left. \frac{\partial \ell}{\partial \eta_i} \right|_{\hat{\eta}} = \delta_i - \exp(\hat{\eta}_i) \sum_{j: t_j \leq t_i} \frac{\delta_j}{\sum_{m: t_m \geq t_j} \exp(\hat{\eta}_m)} \quad (\text{Cox score}) \quad (8)$$

$$W_{ii} = - \left. \frac{\partial^2 \ell}{\partial \eta_i^2} \right|_{\hat{\eta}} = \exp(\hat{\eta}_i) \sum_{j: t_j \leq t_i} \frac{\delta_j}{\sum_{m: t_m \geq t_j} \exp(\hat{\eta}_m)} \quad (\text{negative diagonal Hessian}) \quad (9)$$

After completing the square, the survival log-likelihood becomes

$$\ell(\eta) \approx \sum_i \left[-\frac{1}{2} W_{ii} (\eta_i - z_i)^2 \right] + C \quad (10)$$

where the **working response** is

$$z_i = \hat{\eta}_i + \frac{u_i}{W_{ii}} \quad (11)$$

Equation (10) is a weighted Gaussian likelihood with response z_i , precision W_{ii} , and mean $\eta_i = \sum_k l_{ik} \beta_k$.

4 Update for q_{l_k} — Patient Loadings

4.1 Partial Residuals

To isolate the contribution of factor k , define the **genomics partial residual** (residual from all factors *except* k):

$$R_{ij}^{-k} = Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}, \quad \text{so that} \quad Y_{ij} \approx R_{ij}^{-k} + l_{ik} f_{jk} \quad (12)$$

and the **survival partial working response**:

Revision Note

[R2] The 2/12/26 document (Eq. 22) wrote $z_i^{-k} = z_i - \sum_{k' \neq k} l_{ik'} f_{jk'}$, incorrectly using the factor weight $f_{jk'}$ (which carries a feature index j) in a per-sample quantity. The correct definition uses the survival coefficient $\beta_{k'}$:

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'}, \quad \text{so that} \quad z_i \approx z_i^{-k} + l_{ik} \beta_k \quad (13)$$

4.2 Genomics Likelihood Contribution

$$\log P(Y \mid L, F, \tau) = \sum_{ij} \left[-\frac{1}{2} \tau_{ij} \left(R_{ij}^{-k} - l_{ik} f_{jk} \right)^2 \right] \quad (14)$$

$$= \sum_{ij} \left[-\frac{1}{2} \tau_{ij} \left(\cancel{(R_{ij}^{-k})^2} - 2 R_{ij}^{-k} l_{ik} f_{jk} + l_{ik}^2 f_{jk}^2 \right) \right] \quad (15)$$

Collecting terms in l_{ik} and l_{ik}^2 , and taking the expectation over q_f (treating R_{ij}^{-k} as fixed):

$$\text{genomics:} \quad A_{ik}^{\text{gen}} = \sum_j \tau_{ij} \overline{f_{jk}^2}, \quad B_{ik}^{\text{gen}} = \sum_j \tau_{ij} R_{ij}^{-k} \bar{f}_{jk} \quad (16)$$

4.3 Survival Likelihood Contribution

From Eq. (10):

$$\log P(t, \delta \mid L, \beta) \approx \sum_i \left[-\frac{1}{2} W_{ii} \left(z_i^{-k} - l_{ik} \beta_k \right)^2 \right] \quad (17)$$

$$= \sum_i \left[-\frac{1}{2} W_{ii} \left(\cancel{(z_i^{-k})^2} - 2 z_i^{-k} l_{ik} \beta_k + l_{ik}^2 \beta_k^2 \right) \right] \quad (18)$$

Taking the expectation over q_β :

$$\text{survival:} \quad A_{ik}^{\text{surv}} = W_{ii} \overline{\beta_k^2}, \quad B_{ik}^{\text{surv}} = W_{ii} z_i^{-k} \bar{\beta}_k \quad (19)$$

4.4 Combined Coefficients and EBNM Problem

Revision Note

[R3] Equations 32–35 in the 2/12/26 document wrote f_{jk}^2 and β_k^2 where *second moments* $\mathbb{E}_q[f_{jk}^2]$ and $\mathbb{E}_q[\beta_k^2]$ are required. The corrected forms are below.

The total quadratic form in l_{ik} is $-\frac{1}{2}A_{ik} l_{ik}^2 + B_{ik} l_{ik}$, where

$$A_{ik} = \sum_j \tau_{ij} \overline{f_{jk}^2} + W_{ii} \overline{\beta_k^2} \quad (20)$$

$$B_{ik} = \sum_j \tau_{ij} R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k \quad (21)$$

EBNM Problem

For each sample i , solve the EBNM problem with pseudo-observation x_i and pseudo-standard-deviation s_i :

$$x_i = \frac{B_{ik}}{A_{ik}}, \quad s_i = \frac{1}{\sqrt{A_{ik}}} \quad \Rightarrow \quad (\hat{g}_{l_k}, \hat{q}_{l_k}) = \text{EBNM}(\mathbf{x}, \mathbf{s})$$

The EBNM solver returns the fitted prior $\hat{g}_{l_k}$ and the posterior moments: $\bar{l}_{ik} = \mathbb{E}_{\hat{q}}[l_{ik}]$ and $\bar{l}_{ik}^2 = \text{Var}_{\hat{q}}(l_{ik}) + \bar{l}_{ik}^2$.

5 Update for q_{f_k} — Biological Factors

F appears only in the genomics term of the likelihood (not in the survival term), so the update is identical to the standard EBMF derivation.

5.1 Coefficients

Using the same partial residual R_{ij}^{-k} from Eq. (12) and taking the expectation over q_l :

$$A_{jk} = \sum_i \tau_{ij} \bar{l}_{ik}^2, \quad B_{jk} = \sum_i \tau_{ij} R_{ij}^{-k} \bar{l}_{ik} \quad (22)$$

5.2 EBNM Problem

Revision Note

[R4] Equations 47–48 of the 2/12/26 document wrote the EBNM inputs as $x_i = B_{jk}/A_{jk}$ and $s_i^2 = 1/A_{jk}$. The subscript must be j (feature index) because we produce one pseudo-observation *per feature* when updating factor k .

EBNM Problem

For each feature j , solve the EBNM problem with

$$x_j = \frac{B_{jk}}{A_{jk}} = \frac{\sum_i \tau_{ij} R_{ij}^{-k} \bar{l}_{ik}}{\sum_i \tau_{ij} \bar{l}_{ik}^2}, \quad s_j = \frac{1}{\sqrt{A_{jk}}} \Rightarrow (\hat{g}_{f_k}, \hat{q}_{f_k}) = \text{EBNM}(\mathbf{x}, \mathbf{s})$$

6 Update for q_{β_k} — Survival Coefficients

β appears only in the survival term, so we focus on

$$\mathcal{F}(q_\beta, g_\beta) = \mathbb{E}_{q_l, q_\beta} [\log P(t, \delta \mid L, \beta)] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \quad (23)$$

6.1 Expanding Over the Posterior of L

Starting from Eq. (10):

$$\mathbb{E}_{q_l, q_\beta} \left[\sum_i -\frac{1}{2} W_{ii} (z_i - \eta_i)^2 \right], \quad \eta_i = \sum_k l_{ik} \beta_k \quad (24)$$

We need $\mathbb{E}_{q_l} [(\sum_k l_{ik} \beta_k)^2]$. Expanding:

Revision Note

[R5] Equation 62 in the 2/12/26 document wrote $(\sum_k l_{ik} \beta_k)^2 = \sum_k l_{ik}^2 \beta_k + \dots$, missing the exponent on β_k . The correct expansion is:

$$\left(\sum_k l_{ik} \beta_k \right)^2 = \sum_k l_{ik}^2 \beta_k^2 + 2 \sum_{k < k'} l_{ik} \beta_k l_{ik'} \beta_{k'} \quad (25)$$

Taking $\mathbb{E}_{q_l}[\cdot]$ and using mean-field independence ($l_{ik} \perp l_{ik'}$ under q):

$$\mathbb{E}_{q_l} \left[\left(\sum_k l_{ik} \beta_k \right)^2 \right] = \sum_k \bar{l}_{ik}^2 \beta_k^2 + 2 \sum_{k < k'} \bar{l}_{ik} \bar{l}_{ik'} \beta_k \beta_{k'} = \left(\sum_k \bar{l}_{ik} \beta_k \right)^2 + \sum_k \text{Var}_q(l_{ik}) \beta_k^2 \quad (26)$$

The last equality follows from $\bar{l}_{ik}^2 = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2$. Substituting into the objective and re-grouping:

$$\begin{aligned} \mathcal{F}(q_\beta, g_\beta) = & \mathbb{E}_{q_\beta} \left[\sum_i -\frac{1}{2} W_{ii} \left(z_i - \sum_k \bar{l}_{ik} \beta_k \right)^2 \right] \\ & - \mathbb{E}_{q_\beta} \left[\sum_k \frac{1}{2} \left(\sum_i W_{ii} \text{Var}_q(l_{ik}) \right) \beta_k^2 \right] + \sum_k \mathbb{E}_{q_{\beta_k}} \left[\log \frac{g_{\beta_k}(\beta_k)}{q_{\beta_k}(\beta_k)} \right] \end{aligned} \quad (27)$$

6.2 Coordinate-Ascent EBNM Problem for β_k

Revision Note

[R6] The 2/12/26 document derived Eq. (27) but then proposed a generic WLS solver, which is inconsistent with the EBNM approach used for L and F (and with the code, which calls `ebnm()` for each β_k). The correct EBNM formulation is derived here.

Fix all $\beta_{k'}, k' \neq k$. Define the **survival partial working response** for β_k :

$$z_i^{-k} = z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \quad (28)$$

The two terms in Eq. (27) that involve β_k are

$$\text{from term 1: } -\frac{1}{2} \left[\sum_i W_{ii} \bar{l}_{ik}^2 \right] \beta_k^2 + \left[\sum_i W_{ii} z_i^{-k} \bar{l}_{ik} \right] \beta_k \quad (29)$$

$$\text{from term 2: } -\frac{1}{2} \left[\sum_i W_{ii} \text{Var}_q(l_{ik}) \right] \beta_k^2 \quad (30)$$

Combining and using $\overline{l_{ik}^2} = \text{Var}_q(l_{ik}) + \bar{l}_{ik}^2$:

$$A_k = \sum_i W_{ii} \overline{l_{ik}^2} \quad (31)$$

$$B_k = \sum_i W_{ii} z_i^{-k} \bar{l}_{ik} \quad (32)$$

EBNM Problem

For each factor k , solve the EBNM problem with a single pseudo-observation:

$$x_k = \frac{B_k}{A_k} = \frac{\sum_i W_{ii} z_i^{-k} \bar{l}_{ik}}{\sum_i W_{ii} \overline{l_{ik}^2}}, \quad s_k = \frac{1}{\sqrt{A_k}} \quad \Rightarrow \quad (\hat{g}_{\beta_k}, \hat{q}_{\beta_k}) = \text{EBNM}(x_k, s_k)$$

Key remark. Using $\overline{l_{ik}^2}$ in A_k rather than $\bar{l}_{ik}^2$ naturally implements the *error-in-variables* correction from Eq. (26): the posterior uncertainty in L inflates the effective noise, preventing β from over-fitting to uncertain loadings. The prior g_{β_k} is estimated from data by the EBNM solver (no fixed σ^2 required).

6.3 Connection to Weighted Least Squares

Equation (27) (first term) has the form of a weighted regression:

$$z_i = \sum_k \bar{l}_{ik} \beta_k + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, W_{ii}^{-1})$$

so the WLS estimator for β (treating $\bar{L}$ as fixed and ignoring prior and variance-correction terms) is $\hat{\beta} = (\bar{L}^\top W \bar{L})^{-1} \bar{L}^\top W \mathbf{z}$. The coordinate-ascent EBNM update derived above is the natural Bayesian generalisation of this WLS step that: (i) accounts for uncertainty in L via $\bar{l}_{ik}^2$, and (ii) applies an EBNM shrinkage prior on β_k .

7 Update for τ — Noise Precision

The precision update follows Section A.1 of the EBMF paper.

7.1 Objective

The components of the ELBO that involve τ are:

$$\mathcal{F}(\tau) = \mathbb{E}_{q_l, q_f} \left[\sum_{ij} \left(\frac{1}{2} \log \tau_{ij} - \frac{1}{2} \tau_{ij} \left(Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 \right) \right] + C \quad (33)$$

Revision Note

[R7] Equation 83 of the 2/12/26 document wrote $\frac{1}{2} \sum_{ij} [\log \tau_{ij} + \tau_{ij} \bar{R}_{ij}^2]$, which has the wrong sign on the second term. Maximizing $\log \tau + \tau \bar{R}^2$ with respect to $\tau > 0$ gives no finite maximum. The correct expression is:

$$\mathcal{F}(\tau) = \frac{1}{2} \sum_{ij} [\log \tau_{ij} - \tau_{ij} \bar{R}_{ij}^2] + C \quad (34)$$

7.2 Expected Squared Residuals

Revision Note

[R8] Equation 85 of the 2/12/26 document wrote the last term as $\sum_k \bar{l}_{jk}^2 \bar{f}_{jk}^2$, with the subscript jk on the loading. The loading carries a *sample* index i , not j . Corrected below.

Since l and f are distributions under q , we require the expected squared residual. Under the mean-field assumption ($q(L, F) = q(L) q(F)$):

$$\bar{R}_{ij}^2 := \mathbb{E}_{q_l, q_f} \left[\left(Y_{ij} - \sum_k l_{ik} f_{jk} \right)^2 \right] = \left(Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk} \right)^2 + \sum_k \left[\bar{l}_{ik}^2 \bar{f}_{jk}^2 - \bar{l}_{ik}^2 \bar{f}_{jk}^2 \right] \quad (35)$$

The first term is the squared residual evaluated at posterior means; the second term corrects for posterior uncertainty in both L and F (variance inflation).

7.3 Precision Estimators

Setting $\partial \mathcal{F} / \partial \tau_{ij} = 0$ in Eq. (34):

$$\hat{\tau}_{ij} = \frac{1}{\bar{R}_{ij}^2} \quad (36)$$

Typical constrained structures:

$$\text{Constant } (\tau_{ij} = \tau): \quad \hat{\tau} = \frac{NP}{\sum_{ij} \bar{R}_{ij}^2} \quad (37)$$

$$\text{Column-specific } (\tau_{ij} = \tau_j): \quad \hat{\tau}_j = \frac{N}{\sum_i \bar{R}_{ij}^2} \quad (38)$$

8 Complete Algorithm Summary

Supervised Matrix Factorization — CAVI Algorithm

Inputs: $Y_{n \times p}$, $(t_i, \delta_i)_{i=1}^n$, number of factors K

Initialize: $\bar{L}, \bar{F}$ (e.g., via SVD of Y); $\bar{\beta}$ (e.g., via Cox on initial loadings); $\hat{\tau}$ (e.g., column-wise sample variance)

For $b = 1, 2, \dots$ **until convergence:**

1. **Compute survival working quantities** (one Taylor expansion per outer iteration):

$$\hat{\eta} = \bar{L} \bar{\beta}, \quad (u_i, W_{ii}) = \text{CoxTaylor}(\hat{\eta}_i, t_i, \delta_i), \quad z_i = \hat{\eta}_i + u_i / W_{ii}$$

2. **For** $k = 1, \dots, K$:

(a) **Update** q_{l_k}, g_{l_k} (patient loadings):

$$\begin{aligned} R_{ij}^{-k} &= Y_{ij} - \sum_{k' \neq k} \bar{l}_{ik'} \bar{f}_{jk'}, & z_i^{-k} &= z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \\ A_{ik} &= \sum_j \tau_{ij} \bar{f}_{jk}^2 + W_{ii} \bar{\beta}_k^2, & B_{ik} &= \sum_j \tau_{ij} R_{ij}^{-k} \bar{f}_{jk} + W_{ii} z_i^{-k} \bar{\beta}_k \\ (\bar{l}_{ik}, \bar{l}_{ik}^2) &\leftarrow \text{EBNM}(x_i = B_{ik}/A_{ik}, s_i = 1/\sqrt{A_{ik}}) \end{aligned}$$

(b) **Update** q_{f_k}, g_{f_k} (biological factors):

$$\begin{aligned} A_{jk} &= \sum_i \tau_{ij} \bar{l}_{ik}^2, & B_{jk} &= \sum_i \tau_{ij} R_{ij}^{-k} \bar{l}_{ik} \\ (\bar{f}_{jk}, \bar{f}_{jk}^2) &\leftarrow \text{EBNM}(x_j = B_{jk}/A_{jk}, s_j = 1/\sqrt{A_{jk}}) \end{aligned}$$

(c) **Update** q_{β_k}, g_{β_k} (survival coefficients):

$$\begin{aligned} z_i^{-k} &= z_i - \sum_{k' \neq k} \bar{l}_{ik'} \bar{\beta}_{k'} \\ A_k &= \sum_i W_{ii} \bar{l}_{ik}^2, & B_k &= \sum_i W_{ii} z_i^{-k} \bar{l}_{ik} \\ (\bar{\beta}_k, \bar{\beta}_k^2) &\leftarrow \text{EBNM}(x_k = B_k/A_k, s_k = 1/\sqrt{A_k}) \end{aligned}$$

3. **Update** $\hat{\tau}$ (noise precision, column-specific):

$$\bar{R}_{ij}^2 = (Y_{ij} - \sum_k \bar{l}_{ik} \bar{f}_{jk})^2 + \sum_k (\bar{l}_{ik}^2 \bar{f}_{jk}^2 - \bar{l}_{ik}^2 \bar{f}_{jk}^2), \quad \hat{\tau}_j = \frac{N}{\sum_i \bar{R}_{ij}^2}$$

9 Errata Table — Summary of Revisions

ID	Location	Error in 2/12/26 doc	Correction
R1	Sec. 1A, model	Wrong dimension subscripts on Y, L, F, E	Fixed in Sec. 2
R2	Eq. 22	z_i^{-k} defined with $f_{jk'}$ instead of $\beta_{k'}$	Eq. (13)
R3	Eqs. 32–35	f_{jk}^2, β_k^2 should be $\mathbb{E}_q[f_{jk}^2], \mathbb{E}_q[\beta_k^2]$	Eqs. (20)–(21)
R4	Eqs. 47–48	EBNM inputs subscripted i (sample); should be j (feature)	Sec. 5
R5	Eq. 62	β_k missing square: should be β_k^2	Eq. (25)
R6	Sec. 6	β update framed as generic WLS, EBNM form missing	Eqs. (31)–(32)
R7	Eq. 83	Sign: $\log \tau + \tau \bar{R}^2$ should be $\log \tau - \tau \bar{R}^2$	Eq. (34)
R8	Eq. 85	$\bar{l}_{jk}^2$ has wrong index j ; should be $\bar{l}_{ik}^2$	Eq. (35)

10 Notes on the Current R Implementation

The code in `Supervised_Bayesian_MF.R` correctly implements the EBNM-based coordinate ascent for L, F, β , and τ . Two approximations in the code are worth documenting:

Orthogonalization resets second moments (line 217). Every 10 iterations the code applies an SVD rotation to $\bar{L}$ for identifiability, then resets `EL2 <- EL^2` and `EF2 <- EF^2`. This drops the posterior variance, setting $\bar{l}_{ik}^2 \leftarrow \bar{l}_{ik}^2$. The correct second moment is $\bar{l}_{ik}^2 + \text{Var}_q(l_{ik}) \geq \bar{l}_{ik}^2$. The variance is recovered at the next EBNM call, so the algorithm remains consistent but may have a transient underestimate of posterior uncertainty every 10 iterations.

Survival Taylor expansion computed once per outer iteration. The working quantities z_i and W_{ii} are computed once per outer loop (before the inner k -loop), so they use the linear predictor $\hat{\eta}$ from the *start* of each outer iteration. Within the inner loop, as $\bar{L}$ and $\bar{\beta}$ are updated, the true linear predictor changes but z_i and W_{ii} are not refreshed. This is a standard approximation in IRLS-within-VI algorithms and is acceptable; the outer loop restores exactness at each iteration.